

A CLASSIFICATION OF INTEGER-SIDED CYCLIC POLYGONS WITH SEMIPRIME RADIUS

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ABSTRACT. For a positive integer n , let $a(n)$ be the largest number of sides of a simple cyclic polygon of circumradius n whose side lengths are positive integers. We determine $a(pq)$ for all primes $p \leq q$. For a prime $r \equiv 1 \pmod{3}$, let $s_r > 0$ be minimal among the solutions of $s_r^2 + 3u_r^2 = 4r^2$, and put $x_r = \arcsin(s_r/(2r))$. We prove that $a(pq) = 18$ when p, q are odd, $p \equiv q \equiv 1 \pmod{3}$, and $x_p + x_q < \pi/6$; otherwise $a(pq) = 8$ when $p < q$ and $p \equiv q \equiv 1 \pmod{4}$; in every remaining case $a(pq) = 6$. The first condition is equivalent to an explicit integer inequality. The proof combines the basis and splitting theorem for pure geodetic angles of Conway, Radin, and Sadun with ideal-class arguments in imaginary quadratic fields. All remaining finite checks are supplied as exact Maple certificates and independent Python reimplementations.

1. INTRODUCTION

A cyclic polygon is a polygon whose vertices lie on a common circle. We fix the circumradius to be a positive integer, require every side length to be a positive integer, and maximize the number of sides.

For $n \geq 1$ and $1 \leq s \leq 2n$, define

$$(1) \quad \alpha_n(s) = \arcsin\left(\frac{s}{2n}\right) \in (0, \pi/2].$$

Thus $2\alpha_n(s)$ is the minor central angle subtended by a chord of length s .

Definition 1.1. Let $a(n)$ be the largest integer k for which there is a simple cyclic k -gon of circumradius n with positive integer side lengths.

The regular hexagon of side length n shows that $a(n) \geq 6$ for every n . The sequence is recorded as OEIS A398490. Its values at semiprime radii display large jumps: for example,

$$a(49) = 18, \quad a(65) = 8, \quad a(91) = 18, \quad a(361) = 6.$$

The following theorem explains all such behavior.

For a prime $r \equiv 1 \pmod{3}$, choose the solution of

$$(2) \quad s_r^2 + 3u_r^2 = 4r^2$$

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with $s_r > 0$ minimal, and put

$$(3) \quad x_r = \arcsin\left(\frac{s_r}{2r}\right) \in (0, \pi/6).$$

Theorem 1.2 (Semiprime classification). *Let $p \leq q$ be primes. Then*

$$a(pq) = \begin{cases} 18, & p, q \text{ are odd, } p \equiv q \equiv 1 \pmod{3}, \\ & x_p + x_q < \pi/6, \\ 8, & p < q, \quad p \equiv q \equiv 1 \pmod{4}, \quad \text{and the first case fails,} \\ 6, & \text{otherwise.} \end{cases}$$

Moreover, $x_p + x_q < \pi/6$ is equivalent to

$$(4) \quad 3(u_p s_q + u_q s_p) < 3u_p u_q - s_p s_q.$$

In particular,

$$a(pq) \in \{6, 8, 18\}, \quad a(2p) = a(3p) = 6$$

for every prime p .

The proof has three layers. Section 2 translates the geometry into linear relations among half-angles. Sections 3 through 6 use the Conway–Radin–Sadun basis to split those relations into squarefree classes and independent prime components. The ordinary classes share a common four-angle model, while the Gaussian and Eisenstein classes produce the exceptional values 8 and 18. Finally, Section 8 treats $a(2p)$ by a finite exact ideal-class audit.

2. THE GEOMETRIC CRITERION

Lemma 2.1 (Minor- and major-arc criterion). *A multiset of positive integers $\{s_1, \dots, s_k\}$, with $s_i \leq 2n$, is realizable as the side-length multiset of a simple cyclic polygon of radius n if and only if either*

$$(5) \quad \sum_{i=1}^k \alpha_n(s_i) = \pi,$$

or

$$(6) \quad \alpha_n(s_j) = \sum_{i \neq j} \alpha_n(s_i)$$

for some j .

Proof. If every side uses its minor arc, the central angles $2\alpha_n(s_i)$ sum to 2π , giving (5). A simple cyclic polygon can use a major arc for at most one side. If s_j uses it, its central angle is $2\pi - 2\alpha_n(s_j)$, while all other central angles are minor; their sum is 2π exactly when (6) holds. Conversely, arranging the corresponding arcs consecutively around the circle constructs a simple cyclic polygon. $\square$

We call (5) a π -block. A positive relation with sum $\pi/2$ is called a $\pi/2$ -block. Relation (6) is a major-arc relation. A side of length $2n$ is a diameter; in this case the remaining sides form a $\pi/2$ -block.

We shall also use the strict monotonicity

$$(7) \quad \frac{d}{ds} \alpha_n(s) = \frac{1}{\sqrt{4n^2 - s^2}} > 0.$$

Thus a half-angle uniquely determines its side length at fixed radius.

3. GEODETIC ANGLES AND COMPONENT CANCELLATION

For a side s , write

$$(8) \quad 4n^2 - s^2 = du^2,$$

where d is squarefree. Its complementary half-angle

$$\beta_n(s) = \frac{\pi}{2} - \alpha_n(s)$$

satisfies

$$(9) \quad \tan \beta_n(s) = \frac{u}{s} \sqrt{d}.$$

Hence $\beta_n(s)$ is a pure geodetic angle in the terminology of Conway, Radin, and Sadun [1].

We use their basis theorem and splitting theorem. In a fixed squarefree class d , write the unique basis expansion of a pure geodetic angle as

$$(10) \quad \theta = q(\theta)\pi + \sum_r c_r(\theta)\Theta_{r,d},$$

where the sum is finite and the prime-indexed basis angles $\Theta_{r,d}$ are independent modulo rational multiples of π .

Lemma 3.1 (Component cancellation). *Fix d . If*

$$\sum_i m_i \theta_i \in \pi \mathbb{Q}$$

for pure geodetic angles θ_i in class d and integers m_i , then

$$\sum_i m_i c_r(\theta_i) = 0$$

for every prime-basis element $\Theta_{r,d}$ in (10). Consequently, the signed components belonging to distinct split rational primes cancel separately.

Proof. Insert the unique expansions (10). After subtracting the rational multiple of π , uniqueness of the Conway–Radin–Sadun basis forces the coefficient of every $\Theta_{r,d}$ to vanish. $\square$

The splitting theorem further says that, in any rational relation among pure geodetic angles, the restriction to each squarefree class d is itself a rational multiple of π . These two facts will be used throughout without further comment.

For later normalization, put

$$(11) \quad g = \gcd(s, u), \quad A = \frac{s}{g}, \quad B = \frac{u}{g}, \quad M = \frac{2n}{g}.$$

Then

$$(12) \quad \gcd(A, B) = 1, \quad A^2 + dB^2 = M^2.$$

Only primes dividing M may occur as prime components of the angle.

4. PRIME-COMPONENT ATOMS

Assume first that $n = pq$ with distinct odd primes p and q .

Lemma 4.1. *If a variable prime $r \in \{p, q\}$ is inert or ramified in $K = \mathbb{Q}(\sqrt{-d})$, then $r \nmid M$ in (12).*

Proof. If r is inert and $r \mid M$, the prime ideal (r) divides $A + B\sqrt{-d}$, and hence $r \mid A, B$, contrary to coprimality. If r is ramified and $r \mid M$, then $r \mid d$. If $r \nmid A$, the valuation of $A^2 + dB^2$ at r is zero; if $r \mid A$, then $r \nmid B$ and squarefreeness of d gives valuation one. Both alternatives contradict the even valuation in M^2 . $\square$

For a split prime r , write $(r) = \mathfrak{r}\bar{\mathfrak{r}}$. If $r \mid M$, the signed exponent difference, divided by two, belongs to $\{-1, 0, 1\}$. Thus every primitive side has a normalized variable-component vector

$$(v_p, v_q) \in \{-1, 0, 1\}^2.$$

Every rational angle relation has total component vector zero by Lemma 3.1.

Lemma 4.2 (Minimal component atoms). *Every finite zero-sum multiset of vectors in $\{-1, 0, 1\}^2$ is a disjoint union of minimal zero-sum multisets, and every minimal nonzero one has length at most 4.*

For a repeated prime $p = q$, the corresponding one-dimensional component set is contained in $\{-2, -1, 0, 1, 2\}$, and every minimal nonzero zero-sum multiset has length at most 3.

Proof. Repeatedly remove a minimal nonempty zero-sum submultiset. It remains to bound the length of one minimal atom J of nonzero vectors in

$$V = \{-1, 0, 1\}^2 \setminus \{(0, 0)\}.$$

If J contains opposite vectors v and $-v$, this pair is already a zero-sum submultiset, so minimality forces $J = \{v, -v\}$.

Assume that no opposite pair occurs. Group equal vectors and write the resulting support-minimal positive relation as

$$a_1 v_1 + \cdots + a_r v_r = 0, \quad a_i \in \mathbb{Z}_{>0},$$

with distinct directions v_i . The origin lies in the convex hull

$$0 \in \text{conv}\{v_1, \dots, v_r\}.$$

By Caratheodory's theorem in dimension two, it lies in the convex hull of at most three of these directions. Support minimality therefore gives $r \leq 3$. The cases $r = 1$ and $r = 2$ are impossible without a zero vector or an opposite pair, so $r = 3$.

For three noncollinear vectors u, v, w surrounding the origin, the primitive positive relation is obtained from

$$\det(v, w)u + \det(w, u)v + \det(u, v)w = 0$$

by choosing a common orientation, taking absolute values, and dividing by the common gcd. Every determinant has absolute value 1 or 2. A determinant has absolute value 2 only for two diagonal vectors. Two of the three determinants cannot both have absolute value 2: after fixing their common diagonal vector, the other two diagonal vectors are opposites, which

has been excluded. Hence at most one primitive coefficient is 2, and the other two are at most 1. The coefficient sum, which is the length of J , is therefore at most $2 + 1 + 1 = 4$.

In one dimension, apart from opposite pairs, the only primitive minimal relations are $2 - 1 - 1 = 0$ and its negative, so the repeated-prime bound is 3. $\square$

We now make the residual-norm step completely explicit. Let $R = \mathbb{Z}[\sqrt{-d}]$ and $\mathcal{O}_K$ be the ring of integers of $K = \mathbb{Q}(\sqrt{-d})$.

Lemma 4.3 (Odd division in the quadratic order). *For every odd positive integer m ,*

$$(13) \quad m\mathcal{O}_K \cap R = mR.$$

Proof. If $\mathcal{O}_K = R$, there is nothing to prove. Otherwise $d \equiv 3 \pmod{4}$ and

$$\mathcal{O}_K = \mathbb{Z}\left[\frac{1 + \sqrt{-d}}{2}\right], \quad \mathcal{O}_K/R \cong \mathbb{Z}/2\mathbb{Z}.$$

Let $z \in m\mathcal{O}_K \cap R$, so $z = m\eta$ for some $\eta \in \mathcal{O}_K$. In the quotient $\mathcal{O}_K/R$,

$$0 = [z] = m[\eta].$$

Multiplication by the odd integer m is an automorphism of the group of order two. Hence $[\eta] = 0$, so $\eta \in R$ and $z \in mR$. The reverse inclusion is immediate. $\square$

Lemma 4.4 (Small residual norm). *Let J be a minimal component atom of length $m \leq 4$ for distinct odd primes p, q . Multiply the primitive side elements belonging to J and cancel the equal prime-ideal exponents above p and q . The resulting element $\xi \in R$ satisfies*

$$(14) \quad N(\xi) = 2^{2k} \quad \text{for some } 0 \leq k \leq m \leq 4.$$

In the repeated-prime case the same conclusion holds with $m \leq 3$.

Proof. For each side let $z_j = A_j + B_j\sqrt{-d} \in R$, with $N(z_j) = M_j^2$ and $M_j \mid 2pq$. Put $z = \prod_{j \in J} z_j \in R$. Component cancellation says that the valuations of the two conjugate prime ideals above p are equal, say A , and likewise those above q are equal, say B . Hence

$$(z) = (p)^A (q)^B \mathfrak{a}$$

for an integral ideal $\mathfrak{a}$, and therefore

$$z = p^A q^B \eta$$

for some $\eta \in \mathcal{O}_K$. Since $p^A q^B$ is odd and $z \in R$, Lemma 4.3 gives $\eta \in R$.

After removing the p - and q -parts, only prime factors above 2 remain. Each $M_j \mid 2pq$ contains at most one factor 2, so

$$N(\eta) = 2^{2k}$$

with $k \leq |J| = m$. Renaming η as ξ proves the claim. $\square$

If the residual element is nonreal, divide its two coefficients by their gcd. Since its norm is supported only at 2, this gives a primitive positive solution

$$(15) \quad x^2 + dy^2 = L^2, \quad L \in \{1, 2, 4, 8, 16\}.$$

The exact elementary enumeration gives, apart from the separately treated Gaussian class $d = 1$,

$$(16) \quad d \in \{3, 7, 15, 23, 31, 39, 55, 87, 231, 247, 255\}.$$

For a repeated prime, $L \leq 8$ and the non-Gaussian list reduces to

$$(17) \quad d \in \{3, 7, 15, 39, 55\}.$$

These finite lists are reproduced by both the Maple and Python certificates.

5. THE ORDINARY SQUAREFREE CLASSES

Every d in (16) other than 3 satisfies $d \equiv 7 \pmod{8}$. For odd n , equation (8) then forces s and u to be even. After division by 4, the primitive norm is odd, so the individual angle has no prime-2 basis component. The same conclusion holds for any ordinary class $d \neq 1, 3$: if a nonzero prime-2 component existed, Lemma 4.4 would place d in the finite list.

The only roots of unity in $\mathbb{Q}(\sqrt{-d})$ for $d \neq 1, 3$ are ± 1 . Consequently, a rational restriction in an ordinary class is an integral multiple of

$$H = \frac{\pi}{2}.$$

A minimal atom with $k \leq 4$ sides has original-angle sum $kH - m\pi$. Positivity leaves only

$$(18) \quad k = 3, \quad \sum \alpha = H, \quad \text{or} \quad k = 4, \quad \sum \alpha = \pi.$$

Thus every ordinary class has density at most three sides per H , or one side per unit $U = \pi/6$.

For completeness we record the sharper common four-angle model used for major arcs. Up to conjugation, the nonzero exponent pairs are $(1, 0)$, $(0, 1)$, $(1, 1)$, and $(1, -1)$. Choose their two independent components so that $0 < X \leq Y < H$. The possible non-diameter half-angles are

$$(19) \quad H - X, \quad H - Y, \quad H - Y + X, \quad |H - X - Y|.$$

If $X + Y < H$, the coefficient equations show that no nonempty positive rational block exists. If $X + Y > H$, write the multiplicities in the order (19) as $m_1, \dots, m_4$. Component cancellation gives

$$m_1 = m_3 + m_4, \quad m_2 = m_4 - m_3, \quad 0 \leq m_3 \leq m_4.$$

The number of angles is $m_3 + 3m_4$, while their rational sum in units of H is $m_3 + m_4$. Hence

$$(20) \quad L_H \leq 3, \quad L_\pi \leq 6.$$

Writing an angle as $cH + aX + bY$, the functional

$$3c + 2a + 2b$$

is at least 1 on all four active vectors and at most 3 on a target vector. Therefore an ordinary major-arc relation has at most four sides.

6. THE GAUSSIAN AND EISENSTEIN CLASSES

6.1. **The Gaussian class** $d = 1$. Put $H = \pi/2$. Suppose first that $p < q$ and both primes are congruent to 1 (mod 4). Let x, y be their two independent small Gaussian angles, ordered so that $0 < y < x < \pi/4$. The complete non-diameter angle list is

$$x - y, y, H - x - y, x, H - x, x + y, H - y, H - x + y.$$

Represent an angle by (c, a, b) if it equals $cH + ax + by$. Every vector in this list satisfies

$$(21) \quad 4c + 2a + b \geq 1.$$

In a rational block, the total x - and y -coefficients vanish. Therefore a sum tH containing L sides satisfies $L \leq 4t$, and in particular

$$(22) \quad L_\pi \leq 8.$$

The bound is attained because

$$(23) \quad (x - y) + y + y + (H - x - y) = H.$$

Doubling gives an octagon.

If

$$s_p^2 + u_p^2 = 4p^2, \quad s_q^2 + u_q^2 = 4q^2,$$

are oriented so that $x > y$, the corresponding side multiset at radius pq is

$$(24) \quad \left\{ \left(\frac{|s_p u_q - u_p s_q|}{2} \right)^{[2]}, (ps_q)^{[4]}, \left(\frac{u_p u_q - s_p s_q}{2} \right)^{[2]} \right\}.$$

Gaussian multiplication proves integrality. For $p = q$, the possible one-variable coefficient pairs are $(2, 0)$, $(-1, 1)$, $(1, 0)$, and $(-2, 1)$ for angles $ax + cH$. Each satisfies $a + 3c \geq 1$, so a π -block has at most six sides. If one variable prime does not split, only a subset of the one-component types occurs and the same bound holds.

6.2. **The Eisenstein class** $d = 3$. Put $U = \pi/6$. Suppose first that $p < q$, both primes are congruent to 1 (mod 3), and write $x = x_p$, $y = x_q$.

If $x + y < U$, the complete non-diameter coefficient list (c, a, b) for angles $cU + ax + by$ is

$$\begin{aligned} \mathcal{E}_+ = \{ & (0, 0, 1), (0, 1, 0), (1, -1, -1), (1, -1, 1), (1, 0, 0), \\ & (1, 1, -1), (1, 1, 1), (2, -1, 0), (2, 0, -1), (2, 0, 1), \\ & (2, 1, 0), (3, -1, -1), (3, -1, 1) \}. \end{aligned}$$

The list follows by multiplying the two Eisenstein prime factors, applying the six units, and folding the arguments into $(0, H]$. Every vector satisfies

$$(25) \quad 3c + a + b \geq 1.$$

Thus a sum tU has at most $3t$ sides, and a π -block has at most 18 sides. Equality is attained by taking six copies of

$$(26) \quad x + y + (U - x - y) = U.$$

The third side length is

$$(27) \quad S = \frac{3u_p u_q - s_p s_q - 3(u_p s_q + u_q s_p)}{4}.$$

The numerator is divisible by 4 by Eisenstein multiplication, and $S > 0$ is exactly criterion (4). The resulting side multiset is

$$(28) \quad \{(qs_p)^{[6]}, (ps_q)^{[6]}, S^{[6]}\}.$$

If $x + y > U$, the complete folded list is

$$\begin{aligned} \mathcal{E}_- = \{ & (-1, 1, 1), (0, 1, 0), (0, 0, 1), (1, 1, -1), (1, 0, 0), \\ & (1, -1, 1), (2, 0, -1), (2, -1, 0), (3, -1, -1), (1, 1, 1), \\ & (2, 1, 0), (2, 0, 1), (3, 1, -1) \}. \end{aligned}$$

Every vector satisfies

$$(29) \quad c + a + b \geq 1.$$

Hence a π -block has at most six sides. If one of p, q does not split in the Eisenstein integers, the available vectors form a subset of the inactive table. For $p = q$, the one-variable specialization gives the same bounds: 18 exactly when $2x_p < U$, and otherwise 6. The boundary $2x_p = U$ is impossible because $\sin^2 x_p$ is rational, whereas $\sin^2(\pi/12) = (2 - \sqrt{3})/4$ is irrational.

Finally, a direct sine expansion gives

$$\sin(U - x_p - x_q) = \frac{3u_p u_q - s_p s_q - 3(u_p s_q + u_q s_p)}{8pq},$$

which proves the equivalence between $x_p + x_q < U$ and (4).

7. MIXTURES AND MAJOR-ARC RELATIONS

The rational restrictions in an ordinary or Gaussian class are integral multiples of $H = \pi/2 = 3U$. In the Eisenstein class they are integral multiples of $U = \pi/6$. The previous sections give the following maximal densities, measured in sides per unit U :

$$(30) \quad \rho_d \leq \begin{cases} 3, & d = 3 \text{ and } x_p + x_q < U, \\ 4/3, & d = 1 \text{ with two distinct split variable primes,} \\ 1, & \text{otherwise.} \end{cases}$$

Indeed, neutral atoms with no variable component have the ordinary bound of at most one side per unit U .

If class d contributes a positive angle A_d and k_d sides, then $k_d \leq \rho_d A_d / U$. Therefore

$$\sum_d k_d \leq \left(\max_d \rho_d \right) \frac{\sum_d A_d}{U}.$$

A positive mixture cannot be denser than its densest participating class. Since a π -block has total angle $6U$, its global maximum is 18 in the active Eisenstein case, 8 in the remaining two-component Gaussian case, and 6 otherwise. The constructions (23) and (26), together with the regular hexagon, show that these bounds are attained.

Lemma 7.1 (No mixed non-diameter major arc). *Every non-diameter major-arc relation is contained in one squarefree class. A diameter target reduces to a $\pi/2$ -block.*

Proof. Write a non-diameter relation as

$$\alpha_* = \sum_j \alpha_j < H,$$

and let the target angle lie in class d_0 . If a non-target class occurs, the restriction to d_0 equals the sum of the positive contributions of all non-target classes, and is therefore positive and strictly smaller than H . Every restriction is rational by the splitting theorem.

If $d_0 \neq 3$, its restriction is an integral multiple of H , impossible in $(0, H)$. If $d_0 = 3$, its restriction is U or $2U$, while every non-target ordinary or Gaussian restriction is a multiple of $H = 3U$; their sum cannot equal U or $2U$. Hence no non-target class occurs. If the target is a diameter, the remaining angles sum to H . $\square$

Inside one class, the earlier functionals give the following major-arc bounds: ordinary classes at most 4, Gaussian at most 4, active Eisenstein at most 10, and inactive Eisenstein at most 4. A diameter followed by an H -block gives at most 10 sides in the active Eisenstein case, 5 in the Gaussian case, and 4 otherwise. Every major-arc bound is therefore below the corresponding minor-arc maximum.

This completes the proof of Theorem 1.2 for odd p, q .

8. THE EVEN SEMIPRIME CASE

It remains to prove $a(2p) = 6$. Let p be odd. The same residual-norm argument leaves only

$$d \in \{1, 3, 7, 15, 23, 31, 39, 55, 87, 231, 247, 255\}.$$

The classes $d = 1$ and $d = 3$ are elementary. For the other ten classes, the relation pattern depends only on the ideal class of a prime ideal above p , up to conjugation.

Lemma 8.1 (Prime-replacement invariance). *Fix an imaginary quadratic field whose unit group is $\{\pm 1\}$, and fix all ideal factors supported above 2 and the ramified primes. Let P, P' be unramified split prime ideals with $[P'] = [P]$. Then the integer relation lattice among the corresponding side angles, modulo rational multiples of π , is the same for P and P' . The same holds for signed major-arc relations.*

Proof. There is $\lambda \in K^\times$ with $P' = (\lambda)P$. If an admissible side type t has ideal factor $A_t P^{e_t} \overline{P}^{f_t}$, then replacing P by P' changes its phase by

$$(e_t - f_t) \operatorname{Arg}(\lambda) \pmod{\pi}.$$

For a signed relation with coefficients m_t , the total change is

$$\operatorname{Arg}(\lambda) \sum_t m_t (e_t - f_t).$$

The sum is exactly the coefficient of the independent p -basis angle and vanishes by Lemma 3.1. The converse follows by using λ^{-1} . Conjugate ideal classes negate all variable components and preserve the relation lattice. $\square$

Reduced positive binary quadratic forms represent ideal classes; see [2, Chapters 2–3]. Therefore one represented unramified prime per reduced-form orbit under conjugation suffices. The exact ideal-class audit performs the following finite steps:

- (a) enumerate all 37 reduced-form orbits in the ten classes;
- (b) choose one represented unramified prime for each orbit;
- (c) enumerate all exact solutions of $s^2 + du^2 = 16p^2$;
- (d) determine the largest π -block, $\pi/2$ -block, and major-arc relation by exact multiplication in $\mathbb{Q}(\sqrt{d}, i)$;
- (e) separately check all 19 ramified pairs $p \mid d$ and the special $d = 3$ cases.

It finds no violation of

$$(31) \quad L_\pi \leq 6, \quad L_{\pi/2} \leq 3, \quad L_{\text{major}} \leq 4.$$

The regular hexagon gives equality. The remaining case $p = 2$, namely $n = 4$, is checked directly. Thus $a(2p) = 6$ for every prime p .

9. EXACT CERTIFICATES AND REPRODUCIBILITY

The ancillary directory contains the strengthened exact Maple certificate for the odd structural audit, the final even ideal-class audit, an independent Maple cross-audit comparing the recursive maxima with a separate finite Cartesian enumeration, and exact Python reimplementations of the odd and even audits. All proof certificates use exact integer arithmetic and exact multiplication in the basis

$$1, \quad \sqrt{d}, \quad i, \quad i\sqrt{d}.$$

No floating-point trigonometric comparison is used.

For a fixed ordered list $V = (v_1, \dots, v_h)$ of distinct phase vectors, the block recursion processes the types in order and tries multiplicities $0, 1, 2, \dots$ for each type. It prunes only after the exact accumulated phase has crossed the target π , $\pi/2$, or the distinguished major-arc target. Since every remaining angle is positive, no extension of a pruned node can return to equality. Consequently every admissible multiplicity vector is visited exactly once. The separate cross-audit independently derives a finite upper bound for each multiplicity and exhausts the resulting Cartesian box; it agrees with the recursive maximum in every case. The monotonicity (7) ensures that a target half-angle corresponds to a unique side length.

The reproduced outputs are:

Audit	Exact result
Odd semiprime	11 candidate classes; all functional checks passed; the inactive ordinary chamber was excluded by a bound-free linear certificate.
Even semiprime	37 unramified form orbits; 19 ramified pairs; no violations of (31).
Maple cross-audit	Recursive and Cartesian maxima agreed in every case; no bound violations occurred.
Independent Python	Both odd and even audits reproduced the same classes, counts, and bounds.

A separate end-to-end numerical validation, not used as a logical step in the proof, gave

$$a(411) = 6, \quad a(422) = 6, \quad a(445) = 8, \quad a(469) = 18, \quad a(529) = 6.$$

10. EXAMPLES AND CONSEQUENCES

Example 10.1. For $p = 7$ and $q = 13$,

$$(s_7, u_7) = (2, 8), \quad (s_{13}, u_{13}) = (1, 15).$$

Formula (27) gives $S = 61$. Thus

$$\{7^{[6]}, 26^{[6]}, 61^{[6]}\}$$

is the side multiset of an eighteen-gon of radius 91, and $a(91) = 18$.

Example 10.2. For $p = 5$ and $q = 13$, the Gaussian construction gives

$$\{32^{[2]}, 50^{[4]}, 66^{[2]}\},$$

so $a(65) = 8$.

Corollary 10.3. For a prime p ,

$$a(p^2) = 18$$

exactly when $p \equiv 1 \pmod{3}$ and $2x_p < \pi/6$; otherwise $a(p^2) = 6$. In particular,

$$a(49) = 18, \quad a(169) = 18, \quad a(361) = 6.$$

11. FURTHER QUESTIONS

The semiprime classification suggests several natural extensions. Which values can occur for $a(n)$ when n has three distinct prime factors? Can the finite-component method classify $a(mp)$ for other fixed integers m ? More generally, how rapidly can $a(n)$ grow along integers with a prescribed number of prime factors?

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